

Port Security

- Topics covered:
 - Intro to Port Security
 - Why Use Port Security
 - Port Security Configuration

Intro to Port Security

What is Port Security?

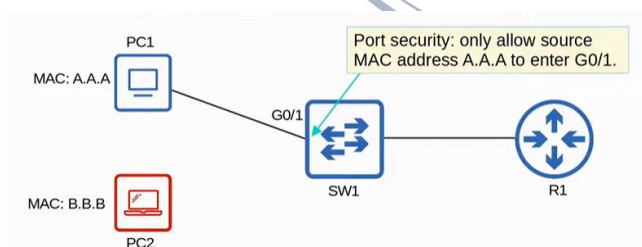
Port security is a Cisco switch security feature that controls which source MAC address(es) are allowed to enter a switchport.

- It is configured on a per-interface basis.
- The terms **port** and **interface** mean the same thing in this context.
- If a frame with an unauthorized source MAC address enters the interface, the switch takes an action.
- By default, the action is to place the interface into an **err-disabled** state.

When an interface becomes err-disabled:

1. It is effectively shut down.
2. The interface no longer sends or receives traffic.

Example: Authorized MAC Address



Port Security on G0/1:

Only allow source MAC address A.A.A

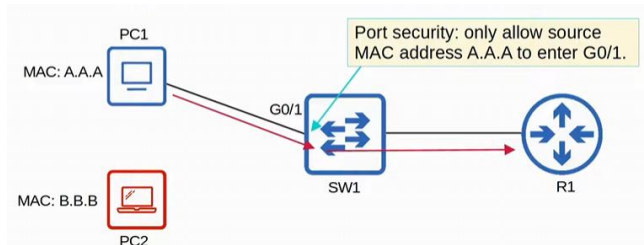
PC1 is connected to SW1's G0/1 interface.

- PC1 MAC address = A.A.A
- Port security is configured to allow only A.A.A on G0/1.

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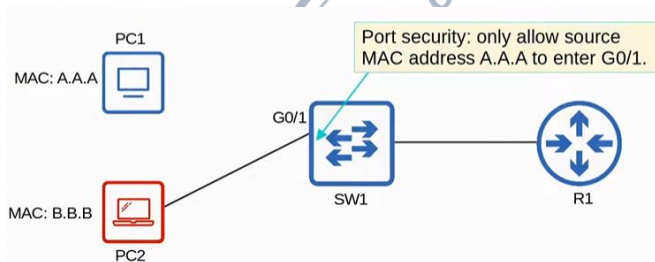
When PC1 sends a frame:

1. SW1 checks the source MAC address.
2. The source MAC is A.A.A.
3. The frame is forwarded normally.



Example: Unauthorized Device Connected

The user disconnects PC1 and connects a personal laptop (PC2).



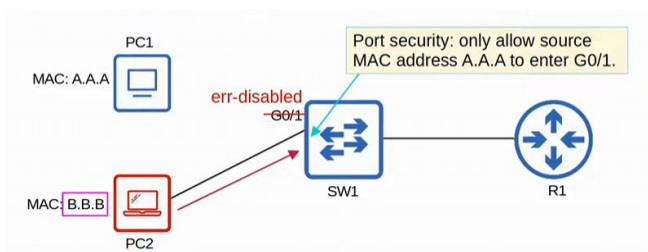
Port Security on G0/1:

Only allow source MAC address A.A.A

When PC2 sends a frame:

1. SW1 checks the source MAC address.
2. The source MAC is B.B.B.
3. Only A.A.A is authorized.
4. SW1 detects a port security violation.
5. G0/1 is placed into an err-disabled state.

Result

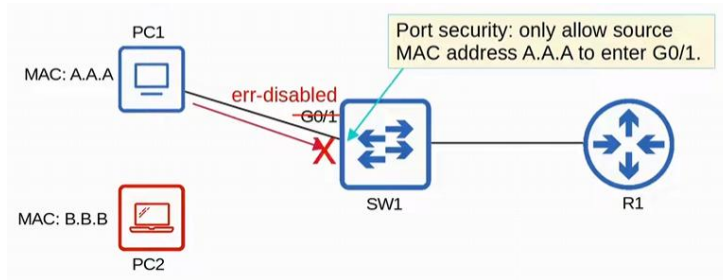


The interface will not send or receive data until it is enabled again.

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Reconnecting the Original PC

The user disconnects PC2 and reconnects PC1.



G0/1 = err-disabled

Even though PC1 has the authorized MAC address:

1. The interface remains err-disabled.
2. PC1 cannot communicate on the network.

There are two methods to re-enable an interface disabled by port security, which are covered later.

Default Port Security Behaviour

When port security is enabled with default settings:

Only **one MAC address** is allowed.

The allowed MAC address can be:

1. Manually configured, or
2. Dynamically learned.

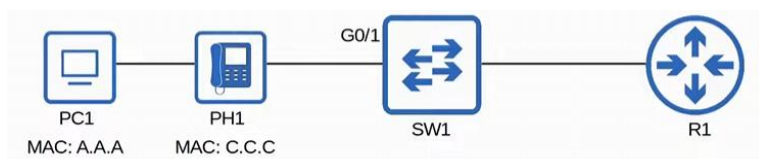
If no MAC address is configured manually:

1. The switch learns the first source MAC address that enters the interface.
2. That MAC address becomes authorized.
3. All other MAC addresses become unauthorized.

Allowing Multiple MAC Addresses:

Sometimes more than one MAC address must be allowed.

Example: IP Phone and PC



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In this topology:

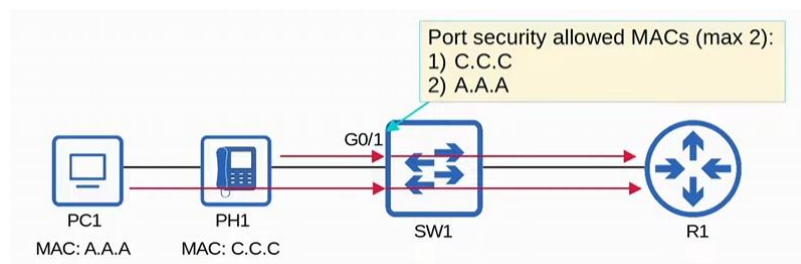
- Phone1 is connected directly to SW1.
- PC1 is connected through Phone1.
- Both devices send traffic using their own MAC addresses.

The default limit of one MAC address is not sufficient.

Configuring Two Allowed MAC Addresses:

SW1 G0/1 is configured to allow two MAC addresses.

The MAC addresses are learned dynamically.



Process:

1. Phone1 sends a frame.

- SW1 learns MAC address C.C.C.
- Frame is forwarded.

2. PC1 sends a frame.

- SW1 learns MAC address A.A.A.
- Frame is forwarded.

3. The maximum number of allowed MAC addresses is now reached.

If another device connects and sends a frame:

1. Its MAC address is not authorized.
2. SW1 shuts down the interface.

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Key Port Security Concepts

1. Number of Allowed MAC Addresses

By default: Maximum allowed MAC addresses = 1

This value can be increased.

2. MAC Address Learning Methods

Allowed MAC addresses can be:

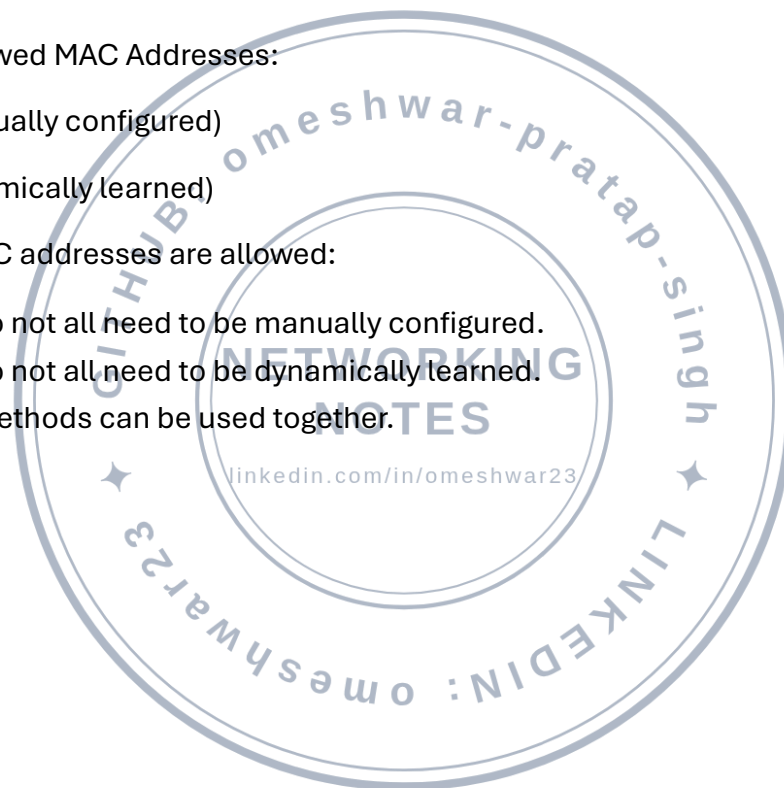
1. Manually configured.
2. Dynamically learned.
3. A combination of both.

Example: Allowed MAC Addresses:

1. C.C.C (manually configured)
2. A.A.A (dynamically learned)

If multiple MAC addresses are allowed:

- They do not all need to be manually configured.
- They do not all need to be dynamically learned.
- Both methods can be used together.



Why Use Port Security?

What is The Use of Port Security?

Port security helps network administrators control which devices can access the network.

Benefits:

- Prevents unauthorized devices from simply being plugged into a switchport and gaining network access.
- Restricts which MAC addresses are permitted.
- Limits how many MAC addresses can use an interface.

Limitation: MAC Address Spoofing

MAC address spoofing is easy to perform.

A device can be configured to send frames using a different source MAC address.

Because of this:

- Port security is not a perfect security solution.
- Administrators should be aware of this limitation.

Protection Against Large Numbers of MAC Addresses

Limiting the number of MAC addresses allowed on an interface can help protect against attacks that generate many MAC addresses.

Potential effects of such attacks include:

- Exhausting DHCP address pools.
- Filling the switch MAC address table.

When the MAC address table becomes full:

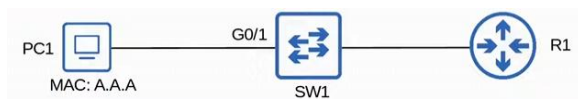
- The switch cannot learn new MAC addresses.
- The switch floods received packets.

Port security can help prevent these situations by restricting how many MAC addresses are permitted on an interface.

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Basic Port Security Configuration

Initial Topology:



Port security is configured directly on the interface.

Attempting to Enable Port Security:

```
SW1(config)#interface g0/1
SW1(config-if)#switchport port-security
Command rejected: GigabitEthernet0/1 is a dynamic port.

SW1(config-if)#do show int g0/1 switchport
Name: Gi0/1
Switchport: Enabled
Administrative Mode: dynamic auto
Operational Mode: static access
! [output omitted]
```

Explanation

The command is rejected because:

The interface administrative mode is **dynamic auto**.

Port security can be enabled on:

- Static access ports
- Static trunk ports

Allowed:

1. switchport mode access
2. switchport mode trunk

Not allowed:

1. switchport mode dynamic auto
2. switchport mode dynamic desirable

Configuring a Static Access Port:

```
SW1(config-if)#switchport mode access

SW1(config-if)#do show int g0/1 switchport
Name: Gi0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
```

The administrative mode is now static access, so the switchport port-security command should work.

Explanation

The interface is now statically configured as an access port.

1. Administrative Mode = static access
2. Port security requirements are now met.

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Enabling Port Security:

```
SW1(config-if)#switchport port-security
SW1(config-if)#
```

The command works, so port security is now enabled on G0/1.

Explanation

The command is accepted.

Port security is now enabled on G0/1.

When only this command is entered:

1. Port security is enabled.
2. All default settings are used.

Viewing Port Security Status

The command below displays the current port security configuration and status of an interface.

```
SW1#show port-security interface g0/1
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 0
Configured MAC Addresses : 0
Sticky MAC Addresses   : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0
```

Default Settings Explained:

Port Security: Enabled.

Port Status: Secure-up.

- Port security is enabled.
- The interface is operational.

Violation Mode: Shutdown.

- This is the default violation mode.
- An unauthorized frame causes the interface to enter the err-disabled state.

Aging Time: 0 minutes.

- Secure MAC addresses never age out.
- No aging timer is active.

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After PC1 Sends Traffic:

PC1 sends a ping to R1.



```
SW1#show port-security interface g0/1
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 0
Last Source Address:Vlan : 000a.000a.000a:1
Security Violation Count : 0
```

Two fields changed:

Total MAC Addresses: 1

- SW1 learned PC1's MAC address.
- The maximum allowed MAC addresses is 1.
- No additional MAC addresses can be learned on the interface.

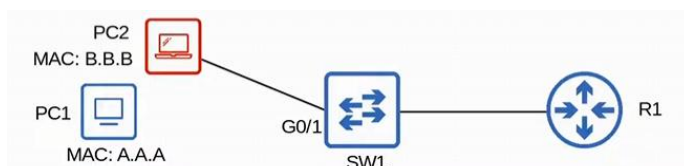
Last Source Address

Last Source Address:Vlan : 000a.000a.000a:1

- Updated to PC1's MAC address.
- VLAN is 1 (default VLAN).

Port Security Violation:

PC1 is disconnected and PC2 is connected.



Previously Learned Secure MAC: A.A.A

PC2 sends traffic using MAC address B.B.B.

Because SW1 already learned A.A.A as the secure MAC address, a violation occurs.

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Result: What Changed?

```
SW1#show port-security interface g0/1
Port Security          : Enabled
Port Status            : Secure-shutdown
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 0
Configured MAC Addresses : 0
Sticky MAC Addresses   : 0
Last Source Address:Vlan : 000b.000b.000b:1
Security Violation Count : 1
```

```
SW1#show interfaces status

```

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	1	auto	auto	unknown
Gi0/1		err-disabled	1	auto	auto	unknown

PC2
MAC: B.B.B

PC1
MAC: A.A.A

G0/1 SW1

R1

Port Status: Secure-up → Secure-shutdown & The interface has been shut down by port security.

Total MAC Addresses 1 → 0

The dynamically learned secure MAC address was cleared when the interface shut down.

Last Source Address: 000b.000b.000b:1

This is PC2's MAC address.

Security Violation Count: 0 → 1 [linkedin.com/in/omeshwar23](https://www.linkedin.com/in/omeshwar23)

One port security violation occurred.

Manually Re-enabling the Interface:

Before entering any commands:

1. Step 1: Disconnect the unauthorized device.
2. Step 2: Re-enable the interface.

```
SW1(config)#interface g0/1
SW1(config-if)#shutdown
SW1(config-if)#no shutdown
```

The interface is administratively shut down and then brought back up.

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Verify the Result

```
SW1#show port-security interface g0/1
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 0
Configured MAC Addresses : 0
Sticky MAC Addresses   : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0
```

1. Port Status returns to Secure-up.
2. Last Source Address is cleared.
3. Security Violation Count is reset.

With the default Shutdown violation mode, the violation counter should not exceed 1 because it is reset when the interface is re-enabled.

ErrDisable Recovery

ErrDisable Recovery automatically re-enables interfaces that have entered the err-disabled state.

Many events can cause an interface to become err-disabled.

Viewing Recovery Settings:

```
SW1#show errdisable recovery
ErrDisable Reason      Timer Status
-----
arp-inspection         Disabled
bpduguard              Disabled
channel-misconfig (STP) Disabled
dhcp-rate-limit        Disabled
dtp-flap               Disabled
! [output omitted due to length]
psecure-violation      Disabled
security-violation     Disabled
sfp-config-mismatch    Disabled
storm-control          Disabled
udld                   Disabled
unicast-flood          Disabled
vmps                   Disabled
psp                    Disabled
dual-active-recovery    Disabled
evc-lite input mapping fa Disabled
Recovery command: "clear" Disabled

Timer interval: 300 seconds

Interfaces that will be enabled at the next timeout:
```

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Explanation

The output contains:

- ErrDisable causes.
- Whether automatic recovery is enabled for each cause.

By default:

psecure-violation: Disabled

Recovery is disabled for all causes.

Default Recovery Timer

Every 5 minutes (by default), all err-disabled interfaces will be re-enabled **if err-disable recovery has been enabled for the cause of the interface's disablement.**

Timer interval: 300 seconds = 5 minutes.

Err-disabled interfaces are automatically re-enabled only if recovery is enabled for the specific cause.

Enabling Recovery for Port Security Violations

```
SW1(config)#errdisable recovery cause psecure-violation
SW1(config)#errdisable recovery interval 180
```

Command	Purpose
errdisable recovery cause psecure-violation	Enable recovery for port-security violations
errdisable recovery interval 180	Set recovery timer to 180 seconds

Verification

```
SW1#show errdisable recovery
ErrDisable Reason    Timer Status
-----
! [output omitted due to length]
psecure-violation    Enabled
! [output omitted due to length]

Timer interval: 180 seconds

Interfaces that will be enabled at the next timeout:

Interface    Errdisable reason    Time left(sec)
-----
Gi0/1        psecure-violation    149
```

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1. Recovery is enabled for port-security violations.
2. Recovery interval is 180 seconds.
3. G0/1 will automatically be re-enabled at the next timeout.
4. 149 seconds remain before recovery.

Critical Requirement

Always remove the unauthorized device first.

Correct Procedure

1. Disconnect the unauthorized device.
2. Re-enable the interface manually, or
3. Allow ErrDisable Recovery to re-enable it automatically.

If the Unauthorized Device Is Not Removed

Manually Configured Secure MAC Address:

When another frame arrives from the unauthorized device:

- The interface becomes disabled again.

Dynamically Learned Secure MAC Address:

When the interface is disabled:

- The learned secure MAC address is removed.

After the interface is re-enabled:

- The unauthorized device's MAC address may become the newly learned secure MAC address.

This is not desirable.

Always disconnect the unauthorized device first.

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Port Security Violation Modes

Port security supports three violation modes.

Comparison:

Feature	Shutdown	Restrict	Protect
Interface Disabled	Yes	No	No
Unauthorized Traffic	Discarded	Discarded	Discarded
Syslog/SNMP Messages	One message when disabled	Message for every violation	No messages
Violation Counter	Set to 1	Incremented for every violation	Not incremented

1. Shutdown Mode (Default)

Behaviour

- Interface enters the err-disabled state.
- The port is effectively shut down.

Notifications

- Generates a Syslog and/or SNMP message.
- Indicates that port security disabled the interface.

After the interface is disabled:

- Additional unauthorized traffic does not generate more messages.
- Only one notification is generated.

Violation Counter

Security Violation Count = 1

When the interface is re-enabled:

Security Violation Count = 0

2. Restrict Mode

Behaviour

- Unauthorized traffic is discarded.
- The interface remains operational.
- Authorized devices continue using the interface normally.

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Notifications

For every unauthorized frame:

- Syslog message generated.
- SNMP message generated.

Violation Counter

For every unauthorized frame:

Security Violation Count +1

3. Protect Mode

Behaviour

- Unauthorized traffic is discarded.
- The interface remains operational.

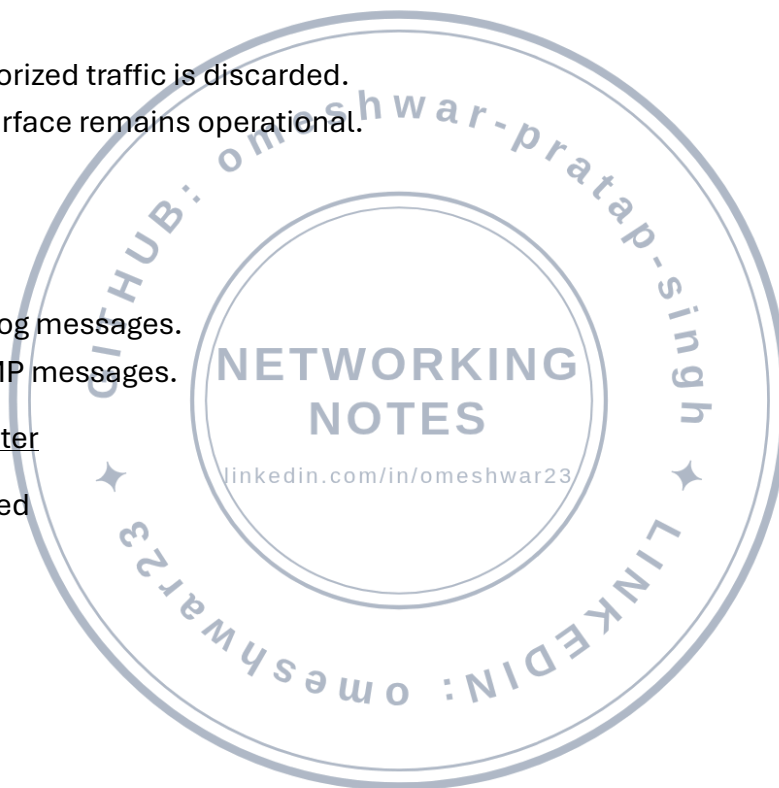
Notifications

None.

- No Syslog messages.
- No SNMP messages.

Violation Counter

Not incremented



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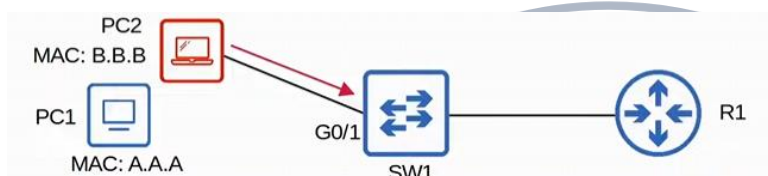
Restrict Violation Mode

Previously, the default **shutdown** mode was demonstrated. Now the focus moves to **restrict** mode.

```
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security mac-address 000a.000a.000a
SW1(config-if)#switchport port-security violation restrict
```

- Port security is enabled.
- PC1's MAC address is manually authorized.
- Violation mode is configured as restrict.

Security Violation:



When PC2 sends traffic:

```
*May 23 22:54:09.951: %PORT_SECURITY-2-PSECURE VIOLATION: Security violation occurred, caused by MAC address 000b.000b.000b on port GigabitEthernet0/1.
```

A syslog message is generated for the unauthorized MAC address.

Verification:

```
SW1#show port-security interface g0/1
Port Security                : Enabled
Port Status                  : Secure-up
Violation Mode                : Restrict
Aging Time                   : 0 mins
Aging Type                   : Absolute
SecureStatic Address Aging   : Disabled
Maximum MAC Addresses        : 1
Total MAC Addresses          : 1
Configured MAC Addresses     : 1
Sticky MAC Addresses         : 0
Last Source Address:Vlan     : 000b.000b.000b:1
Security Violation Count     : 12
```

1. Violation mode is restricted.
2. Port status remains secure-up.
3. Interface is not disabled.
4. Violation counter increases for unauthorized traffic.
5. If the cable is reconnected to PC1, PC1 can still communicate normally because its MAC address is authorized.

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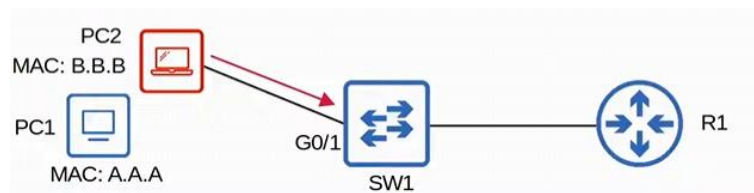
Protect Violation Mode

A fresh port-security configuration is used again.

```
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security mac-address 000a.000a.000a
SW1(config-if)#switchport port-security violation protect
```

1. Port security is enabled.
2. PC1's MAC address is manually authorized.
3. Violation mode is configured as protect.

PC2 sends traffic, but its pings fail.



Verification:

```
SW1#show port-security interface g0/1
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Protect
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 1
Sticky MAC Addresses   : 0
Last Source Address:Vlan : 000b.000b.000b:1
Security Violation Count : 0
```

- Port status remains secure-up.
- Unauthorized traffic is discarded.
- No syslog messages are generated.
- Violation counter remains 0.
- Traffic is silently dropped.

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Violation Mode Summary

Port-security violation modes determine the action taken when an unauthorized frame enters the interface.

Shutdown (Default)

1. Interface enters **err-disabled** state.
2. Syslog and/or SNMP message generated.
3. Violation counter becomes **1**.
4. Counter resets when interface is re-enabled.

Restrict

1. Unauthorized traffic is discarded.
2. Interface remains operational.
3. Syslog and/or SNMP message generated for each violation.
4. Violation counter increments for each unauthorized frame.

Protect

1. Unauthorized traffic is discarded.
2. Interface remains operational.
3. No syslog or SNMP messages.
4. Violation counter does not increase.

Secure MAC Address Aging

MAC addresses that are:

- Dynamically learned, or
- Statically configured

on a port-security-enabled interface are called **secure MAC addresses**.

Default Aging Behaviour:

```
SW1#show port-security interface g0/1
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 1
Configured MAC Addresses : 0
Sticky MAC Addresses    : 0
Last Source Address:Vlan : 000a.000a.000a:1
Security Violation Count : 0
```

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Default Settings

1. Aging time = **0 minutes**
2. Secure static address aging = **Disabled**

This means Secure MAC addresses do **not age out**, they remain until:

- Manually removed, or
- The port is disabled and re-enabled.

Aging Time Configuration

```
switchport port-security aging time <minutes>
```

This command sets the aging timer.

Aging Types

1. Absolute Aging, Default aging type.

```
switchport port-security aging type absolute
```

Behaviour:

1. MAC address is learned.
2. Aging timer starts.
3. Timer continues running even if frames continue arriving from that MAC address.
4. MAC address is removed when the timer expires.
5. If another frame arrives afterward, the MAC can be learned again.

2. Inactivity Aging

```
switchport port-security aging type inactivity
```

Behaviour:

1. MAC address is learned.
2. Aging timer starts.
3. Each received frame resets the timer.
4. As long as traffic continues arriving from that MAC address, it never ages out.

This works like normal MAC address aging.

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Aging Static Secure MAC Addresses

Normally: `switchport port-security mac-address <mac>`

creates a static secure MAC address that does **not** age out.

To allow aging of static secure MAC addresses:

`switchport port-security aging static`

If the MAC ages out:

1. The command is removed from the running configuration.
2. The MAC address is removed from the MAC address table.

Example Configuration

```
SW1(config-if)#switchport port-security aging time 30
SW1(config-if)#switchport port-security aging type inactivity
SW1(config-if)#switchport port-security aging static
```

Verification

```
SW1#show port-security interface g0/1
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Shutdown
Aging Time             : 30 mins
Aging Type             : Inactivity
SecureStatic Address Aging : Enabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 1
Sticky MAC Addresses   : 0
Last Source Address:Vlan : 000a.000a.000a:1
Security Violation Count : 0
```

Changes visible in the output:

1. Aging Time = 30 mins
2. Aging Type = Inactivity
3. SecureStatic Address Aging = Enabled

SHOW PORT-SECURITY

Useful for viewing all interfaces using port security.

```
SW1#show port-security
Secure Port  MaxSecureAddr  CurrentAddr  SecurityViolation  Security Action
          (Count)         (Count)         (Count)
-----
Gi0/1             1             1              0             Shutdown
-----
Total Addresses in System (excluding one mac per port) : 0
Max Addresses limit in System (excluding one mac per port) : 4096
```

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Displays:

1. Interfaces with port security enabled
2. Maximum secure MAC addresses
3. Current secure MAC addresses
4. Violation count
5. Security action

Provides an overall view when multiple interfaces use port security.

Sticky Secure MAC Addresses

Sticky learning allows dynamically learned secure MAC addresses to be automatically added to the running configuration.

Configuration: `switchport port-security mac-address sticky`

How Sticky Learning Works.

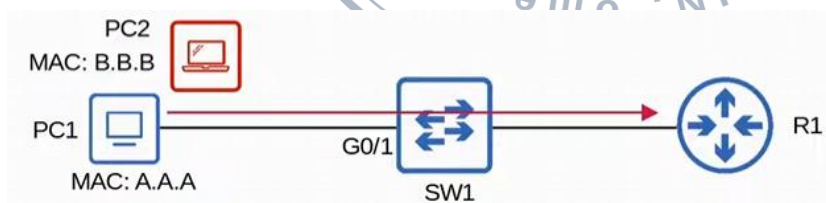
When enabled:

- Dynamically learned secure MAC addresses become sticky secure MAC addresses.
- They are automatically added to the running configuration.

Example

```
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security mac-address sticky
```

And then sent a ping from PC1 to R1.



And then check g0/1 interface.

```
SW1(config-if)#do show running-config interface g0/1
!
interface GigabitEthernet0/1
 switchport mode access
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky 000a.000a.000a
 switchport port-security
 negotiation auto
```

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The following line was added automatically:

```
switchport port-security mac-address sticky 000a.000a.000a
```

It appeared after PC1's frame entered the interface.

Key Point: Sticky MAC addresses provide a way to create static secure MAC addresses without manually configuring the MAC address.

Sticky MAC Address Behaviour

1. Sticky secure MAC addresses never age out.
2. This remains true even when static secure MAC address aging is enabled.
3. Sticky MAC addresses are stored in the **running configuration**.

To make them permanent: write memory

The running configuration must be saved to the startup configuration.

Otherwise, sticky MAC addresses are lost after:

- Reloading the switch
- Power cycling the switch

Converting Between Sticky and Dynamic

When Sticky Learning Is Enabled

1. Existing dynamic secure MAC addresses become sticky secure MAC addresses.
2. They are added to the running configuration.

When Sticky Learning Is Removed

- Sticky secure MAC addresses revert to normal dynamic secure MAC addresses.

Secure MAC Addresses in the MAC Address Table

Secure MAC addresses appear in the MAC address table like other MAC addresses.

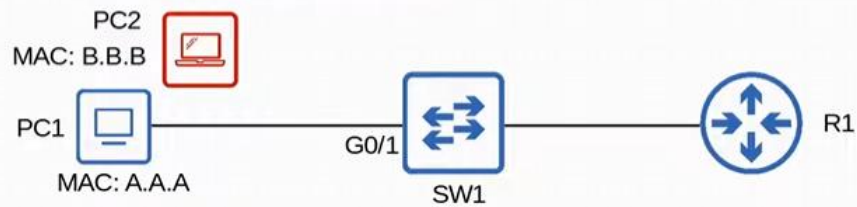
Types

- Sticky secure MAC addresses → **STATIC**
- Static secure MAC addresses → **STATIC**
- Dynamic secure MAC addresses (non-sticky) → **DYNAMIC**

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View Secure MAC Addresses

```
SW1#show mac address-table secure
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
-----
  1     000a.000a.000a  STATIC  Gi0/1
Total Mac Addresses for this criterion: 1
```



In this example:

1. PC1's MAC address appears in the table.
2. Type is **STATIC** because it is a sticky secure MAC address, even though it was originally learned dynamically.



Commands Summary

```
SW1# show mac address-table secure
SW1# show port-security
SW1# show port-security interface interface
SW1# show errdisable recovery
SW1(config)# errdisable recovery cause psecure-violation
SW1(config)# errdisable recovery interval seconds
SW1(config-if)# switchport port-security
SW1(config-if)# switchport port-security mac-address mac-address
SW1(config-if)# switchport port-security mac-address sticky
SW1(config-if)# switchport port-security violation {shutdown | restrict | protect}
SW1(config-if)# switchport port-security aging time minutes
SW1(config-if)# switchport port-security aging type {absolute | inactivity}
SW1(config-if)# switchport port-security aging static
```

